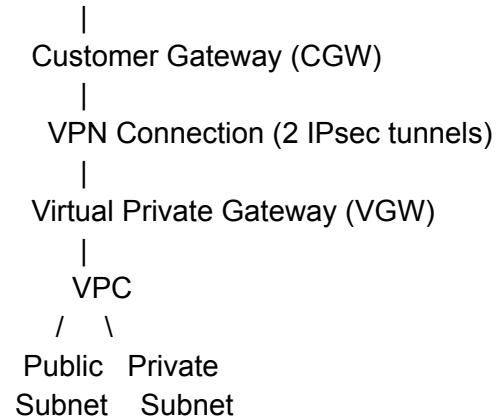

Lab: AWS Site-to-Site VPN – Step-by-Step

Architecture Diagram

[Your Data Center or Simulated Router]



Lab Prerequisites

- AWS account with admin access
- A simulated on-prem router (we'll use **strongSwan** or **VyOS** EC2 for demo)
- VPC with at least one public subnet

♦ Step 1: Create a VPC

- CIDR: `10.0.0.0/16`
- Subnet: `10.0.1.0/24` (public)
- Add Internet Gateway and update route table

♦ Step 2: Launch a Linux EC2 (this will simulate on-prem router)

- AMI: Amazon Linux 2 or Ubuntu
- Name: `onprem-vpn-router`
- Install strongSwan:

```
sudo yum install -y strongswan
```

◆ Step 3: Create a Customer Gateway (CGW)

1. Go to **VPC** → **Customer Gateways**
 2. Click **Create Customer Gateway**
 3. Set:
 - Name: `onprem-cgw`
 - Routing: Static
 - IP address: (Public IP of the EC2 you launched)
 - BGP ASN: (use 65000 or leave default)
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◆ Step 4: Create a Virtual Private Gateway (VGW)

1. Go to **VPC** → **Virtual Private Gateways**
 2. Create VGW with name `my-vgw`
 3. Attach it to your VPC
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◆ Step 5: Create Site-to-Site VPN Connection

1. Go to **VPN Connections** → **Create VPN Connection**
2. Select:
 - Target: **Virtual Private Gateway**
 - Customer Gateway: select existing **onprem-cgw**
 - Routing: Static (**10.0.0.0/16**)
3. Create it and download the configuration file for **strongSwan**

◆ **Step 6: Configure strongSwan on EC2 (simulated on-prem)**

Use the downloaded config or manually edit `/etc/ipsec.conf` and `/etc/ipsec.secrets` to match tunnel IPs and shared secret.

Start the VPN:

```
sudo systemctl restart strongswan
sudo ipsec up vpnconnection
```

◆ **Step 7: Test the VPN**

- Launch another EC2 in the VPC
- Assign internal IP
- From the on-prem EC2, try pinging the EC2 in the VPC

◆ **Step 8: Clean Up Resources**

- Delete VPN connection, VGW, CGW
- Terminate EC2 instances

- Delete VPC